

| COURSE DESCRIPTION            |                                 |               |    |
|-------------------------------|---------------------------------|---------------|----|
| Program                       | Fire Technology and Safety      |               |    |
| Course Title                  | Principles of Safety Management |               |    |
| Course Code                   | 2461                            | Semester      | II |
| Type of Course                | Theory                          | Contact Hours | 60 |
| Teaching Scheme<br>(L: T:P:R) | 4:0:0:0                         | Credits       | 4  |
| CIE Marks                     | 40                              | ESE Marks     | 60 |

### Introduction

*Principles of Safety Management outlines the basic safety principles and practices which serves as a foundation for safety engineering.*

### Course Objectives

|   |                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Foundation of Safety Engineering: To provide a comprehensive understanding of safety engineering principles and the importance of safety in various environments. |
| 2 | Accident Analysis and Prevention: To equip students with the ability to analyze accident causation theories and implement effective prevention                    |
| 3 | Safety Policies and Training Evaluation: To develop skills in assessing the effectiveness of safety policies and training programs within organizations.          |
| 4 | PPE Knowledge: To familiarize students with personal protective equipment (PPE) standards and their practical application in workplace safety.                    |

### Course Pre-requisite

| Topic                      | Course code | Course Title                                                               | Semester |
|----------------------------|-------------|----------------------------------------------------------------------------|----------|
| Basic knowledge in Physics | 2002A       | Engineering Physics for Mechanical, Structural and Industrial Applications | I        |

### Course Outcome

| CO  | Course Outcome                                                                                 | Blooms Taxonomy Level |
|-----|------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | Explain principles of safety engineering and the importance of safety in various environments. | Apply                 |
| CO2 | Explain accident causation theories and find out prevention methods.                           | Understand            |
| CO3 | Explain the effectiveness of safety policies and training programs in the workplace.           | Understand            |
| CO4 | Conduct accident investigations and apply safety performance metrics.                          | Apply                 |

### CO-PO mapping

| COURSE ARTICULATION MATRIX |                  |     |     |     |     |     |     |     |     |      |      |
|----------------------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| Course Outcomes            | PROGRAM OUTCOMES |     |     |     |     |     |     |     |     |      |      |
|                            | PO1              | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
| CO1                        | 3                | -   | -   | -   | -   | 3   | -   | -   | -   | -    | -    |
| CO2                        | 2                | -   | -   | -   | -   | 2   | -   | -   | -   | -    | -    |
| CO3                        | 2                | -   | -   | -   | -   | 2   | 3   | -   | -   | -    | -    |
| CO4                        | 3                | -   | -   | -   | -   | 3   | -   | -   | -   | 2    | -    |
| Total                      | 10               | -   | -   | -   | -   | 10  | 3   | -   | -   | 2    | -    |
| Correlation Level          | 2.5              | -   | -   | -   | -   | 2.5 | 3   | -   | -   | 2    | -    |

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

### Teaching and Assessment Scheme

| Teaching Scheme/Week |   |   |   | Self-learning<br>(SS)/Week | Total<br>Hours/Semester | Credits<br>C | Examination Scheme |     |       |           |     |       |
|----------------------|---|---|---|----------------------------|-------------------------|--------------|--------------------|-----|-------|-----------|-----|-------|
|                      |   |   |   |                            |                         |              | Theory             |     |       | Practical |     |       |
| L                    | T | P | S |                            |                         |              | CIE                | ESE | Total | CIE       | ESE | Total |
| 4                    | 0 | 0 | 0 | 6                          | 60                      | 4            | 40                 | 60  | 100   | 0         | 0   | 0     |

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

### Syllabus - Major Topics

| Module     | Title                            | Major Topics                                                                                                                                                                                                                                                                                                                                                                                                                    | Instructional Hours |   |
|------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---|
|            |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 | L                   | T |
| Module I   | Principles of safety engineering | <ul style="list-style-type: none"> <li>Define key safety terms and concepts related to safety engineering.</li> <li>Explain the context of the safety movement and its significance.</li> <li>Identify various stakeholders' roles in promoting safety within an organization.</li> <li>Analyze different theories of accident causation and their implications for safety practices</li> </ul>                                 | 15                  | 0 |
| Module II  | Accident analysis and prevention | <ul style="list-style-type: none"> <li>Evaluate different methods of accident prevention in the workplace.</li> <li>Assess the importance of safety education and training for employees.</li> <li>Identify barriers to effective communication in safety protocols.</li> <li>Implement best practices for housekeeping using the 5S methodology.</li> </ul>                                                                    | 15                  | 0 |
| Module III | Safety policies and training     | <ul style="list-style-type: none"> <li>Identify various types of personal protective equipment (PPE) and their applications.</li> <li>Analyze PPE standards and their importance in workplace safety.</li> <li>Calculate and interpret safety performance metrics (frequency, severity, incidence, activity rates).</li> <li>Conduct safety inspections and job safety analysis (JSA) to identify potential hazards.</li> </ul> | 15                  | 0 |
| Module IV  | Accident investigation           | <ul style="list-style-type: none"> <li>Describe the key principles and processes involved in accident investigation.</li> <li>Utilize various tools and techniques for data collection during investigations.</li> <li>Analyze accidents using different analytical methods, including MORT and Change Analysis.</li> <li>Conduct case studies to apply investigation techniques and improve safety protocols.</li> </ul>       | 15                  | 0 |

### Detailed Syllabus

| Subtopic                       | Student Learning Outcome                                                                                                                                                                                                                                                                                                     | Mode of delivery (L/T) | Mapped CO | Taxonomy Level | Instructional Hours |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------|----------------|---------------------|
| Module I                       |                                                                                                                                                                                                                                                                                                                              |                        |           |                |                     |
| Introduction                   | Safety - Goals of safety engineering, Need for safety, Safety and productivity                                                                                                                                                                                                                                               | L                      | CO1       | Understand     | 3                   |
| Definitions                    | Accident, Injury, Unsafe act, Unsafe condition, Dangerous occurrence, Reportable accidents                                                                                                                                                                                                                                   | L                      | CO1       | Understand     | 2                   |
| History of Safety Movement     | Overview of the evolution of safety practices                                                                                                                                                                                                                                                                                | L                      | CO1       | Understand     | 2                   |
| Theories of Accident Causation | Analyze different theories of accident causation and their implications for safety practices                                                                                                                                                                                                                                 | L                      | CO1       | Apply          | 4                   |
| Safety Organization            | <ul style="list-style-type: none"> <li>Objectives, types, functions</li> <li>Role of management, supervisors, workmen, unions, government, and voluntary agencies in safety</li> <li>Safety policy</li> <li>Safety Office: responsibilities and authority</li> <li>Safety committee : need, types, and advantages</li> </ul> | L                      | CO1       | Understand     | 4                   |
| Module II                      |                                                                                                                                                                                                                                                                                                                              |                        |           |                |                     |
| Accident Prevention Methods    | Engineering, Education, and Enforcement                                                                                                                                                                                                                                                                                      | L                      | CO2       | Understand     | 3                   |
| Safety Education & Training    | Importance, Various training methods, Effectiveness of training, Behavior-oriented training                                                                                                                                                                                                                                  | L                      | CO2       | Understand     | 3                   |
| Communication                  | Purpose, Barriers to communication                                                                                                                                                                                                                                                                                           | L                      | CO2       | Understand     | 2                   |
| Housekeeping                   | Responsibility of management and employees, Advantages of good housekeeping, 5S of housekeeping                                                                                                                                                                                                                              | L                      | CO2       | Understand     | 3                   |

|                                             |                                                                                                                     |   |     |            |   |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---|-----|------------|---|
| Work Permit System                          | Objectives, Hot work and cold work permits, Typical industrial models and methodology, Entry into confined spaces   | L | CO2 | Understand | 4 |
| Module III                                  |                                                                                                                     |   |     |            |   |
| Personal Protection in the Work Environment | Types of PPEs, Personal protective equipment : respiratory and non-respiratory equipment, Standards related to PPEs | L | CO3 | Understand | 4 |
| Monitoring Safety Performance               | Frequency rate, Severity rate, Incidence rate, Activity rate                                                        | L | CO3 | Understand | 4 |
| Cost of Accidents                           | Computation of costs, Utility of cost data                                                                          | L | CO3 | Understand | 4 |
| Plant Safety Inspection                     | Types, Inspection procedure, Safety sampling techniques, Job safety analysis (JSA), Safety surveys, Safety audits   | L | CO3 | Understand | 4 |
| Module IV                                   |                                                                                                                     |   |     |            |   |
| Accident Investigation                      | Why? When? Where? Who? & How? Basics : Man, Environment & Systems                                                   | L | CO4 | Understand | 4 |
| Process of Investigation                    | Tools, Data collection, Handling witnesses, Case study                                                              | L | CO4 | Understand | 5 |
| Accident Analysis                           | Analytical techniques, System safety, Change analysis, MORT, Multi-events sequencing, TOR                           | L | CO4 | Understand | 6 |

### Suggested Student Activities for Self-Learning

| Week       | Self-Learning description                                                                                                          | Hours |
|------------|------------------------------------------------------------------------------------------------------------------------------------|-------|
| Module I   |                                                                                                                                    |       |
| 1          | Understand the need for safety in daily activities.<br>Prepare short notes on importance of safety                                 | 6     |
| 2          | Study the key terms related to safety.<br>Compare unsafe act and unsafe condition                                                  | 6     |
| 3          | Identify various stakeholders' roles in promoting safety within an organization.<br>Prepare short notes on role of stakeholders    | 6     |
| 4          | Assignment 1- Explain the context of the safety movement and its significance.<br>Understand the history and development of safety | 6     |
| Module II  |                                                                                                                                    |       |
| 5          | Understand various safety training methods<br>Compare safety training methods                                                      | 6     |
| 6          | Identify the barriers to communication.<br>Suggest methods to improve communication                                                | 6     |
| 7          | Assignment 2- Explain in detail about 5S of housekeeping.<br>Practice good housekeeping in labs                                    | 6     |
| 8          | Understand the work permit system<br>Compare types of work permits                                                                 | 6     |
| Module III |                                                                                                                                    |       |
| 9          | Identify various types of PPE's<br>List types of PPE's and its uses                                                                | 6     |
| 10         | Understand various monitoring techniques for safety<br>Prepare short notes                                                         | 6     |
| 11         | Assignment 3-Explain in detail about the standards related to PPE's<br>Prepare short notes                                         | 6     |
| 12         | Understand various inspection techniques<br>Compare the techniques                                                                 | 6     |
| Module IV  |                                                                                                                                    |       |
| 13         | Understand the basics of accident investigation.<br>Observe the various accident analysis techniques                               | 6     |
| 14         | Assignment 4- Prepare a summary of a case study of an accident investigation.<br>Suggest improvements for safety                   | 6     |
| 15         | Understand various accident analysis techniques<br>Prepare short notes                                                             | 6     |

|                           |    |
|---------------------------|----|
| Total Self-Learning hours | 90 |
|---------------------------|----|

### Learning Resources

| TEXT BOOK |                                                      |                         |                                          |
|-----------|------------------------------------------------------|-------------------------|------------------------------------------|
| Sl. No.   | Title                                                | Author                  | Publication                              |
| 1         | Safety management in Industry                        | Krishnan, N.V.          | Jaico Publishing House, New Delhi.(1997) |
| 2         | Accident prevention manual for industrial operations | National Safety Council | National Safety Council, Chicago.(1982)  |

| REFERENCE |                                                |                  |                         |
|-----------|------------------------------------------------|------------------|-------------------------|
| Sl. No.   | Title                                          | Author           | Publication             |
| 1         | Occupational Safety and health                 | David L. Goetsch | PrenticeHall, 2010      |
| 2         | Modern Accident Investigation and Analysis     | Ted S. Ferry     | John Wiley & Sons, 2007 |
| 3         | Occupational Safety Management and Engineering | Willie Hammer    | PrenticeHall, 1989      |
| 4         | Industrial Safety Management                   | Akhil Kumar Das  | PHI,2020                |

| WEB RESOURCES |                                                                                             |                 |
|---------------|---------------------------------------------------------------------------------------------|-----------------|
| Sl. No.       | URL                                                                                         | Modules Covered |
| 1             | <a href="https://share.google/sliyvA5qg3igshT6G">https://share.google/sliyvA5qg3igshT6G</a> | All             |
| 2             | <a href="https://share.google/ZfwMGBdhDIOAJbuoa">https://share.google/ZfwMGBdhDIOAJbuoa</a> | All             |

### Assessment Methodology - Theory

| Mode               | Attendance      | CA1                                 | CA2                                 | CA3                                 | CA4                                 | CA5                     | CA6                     | ESE                   |
|--------------------|-----------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------|-------------------------|-----------------------|
|                    |                 | Self-learning activities (Module 1) | Self-learning activities (Module 2) | Self-learning activities (Module 3) | Self-learning activities (Module 4) | Series Exam (2 modules) | Series Exam (2 modules) | Written Exam (theory) |
| Duration           |                 | 24                                  | 24                                  | 24                                  | 18                                  | 2 hours                 | 2 hours                 | 2.5 hours             |
| Exam mark          |                 | 15                                  | 15                                  | 15                                  | 15                                  | 50                      | 50                      | 60                    |
| Converted to       |                 |                                     |                                     |                                     |                                     | 20                      | 20                      |                       |
| Marks              | 5               | 15                                  |                                     |                                     |                                     | 20                      |                         | 60                    |
| Total              | 40              |                                     |                                     |                                     |                                     |                         |                         | 60                    |
| Tentative schedule | End of semester | By 4th week                         | By 7th week                         | By 11th week                        | By 14th week                        | 8th week                | 15th week               | End of semester       |

### Series Examination Pattern- Theory

| Time    | Module         | Part A | Marks | Part B | Marks | Part C (With choice) | Marks | Total |
|---------|----------------|--------|-------|--------|-------|----------------------|-------|-------|
| 2 hours | Any one module | 2      | 1     | 3      | 3     | 2                    | 7     | 25    |
|         | Any one module | 2      | 1     | 3      | 3     | 2                    | 7     | 25    |
|         | Number         | 4      |       | 6      |       | 4                    |       | 50    |

### End Semester Examination Pattern of Question Paper - Theory

| Duration | Module | TYPE OF QUESTIONS                        |                        |                                                  |                         |                                                          |                         | Total Marks |
|----------|--------|------------------------------------------|------------------------|--------------------------------------------------|-------------------------|----------------------------------------------------------|-------------------------|-------------|
|          |        | Part A<br>(2 Questions from each module) |                        | Part B<br>2 out of 3 questions from each module) |                         | Part C<br>(1 set questions with choice from each module) |                         |             |
|          |        | No. of Questions                         | Marks<br>(1 mark each) | No. of Questions                                 | Marks<br>(3 marks each) | No. of Questions                                         | Marks<br>(7 marks each) |             |

|           |   |   |   |    |    |        |    |    |
|-----------|---|---|---|----|----|--------|----|----|
| 2.5 hours | 1 | 2 | 2 | 3  | 6  | 1 set  | 7  | 15 |
|           | 2 | 2 | 2 | 3  | 6  | 1 set  | 7  | 15 |
|           | 3 | 2 | 2 | 3  | 6  | 1 set  | 7  | 15 |
|           | 4 | 2 | 2 | 3  | 6  | 1 set  | 7  | 15 |
|           |   | 8 | 8 | 12 | 24 | 4 sets | 28 | 60 |

*SITTTR*